21/10/23 M/S

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THIRD SEMESTER B.Tech EXAMINATION, OCTOBER - 2023

PROBABILITY AND VECTOR SPACES

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. **ANSWER ALL QUESTIONS**

		PART-A			**5 X 2=10**
1.	**a.**	Two cards are drawn in succession from a pack of 52 cards. Find the probability that the first is king and the second is queen if the first card is not replaced with the second card.	L-2	CO1	**02**
	b.	If Var(X) = 9 and E(X)=4, then find $E(X^2)$.	L-1	CO2	**02**
	c.	Let V be the set of all ordered pairs (x, y) where x, y are real numbers. Let u= (x_1, y_1) and v= (x_2, y_2) be two elements in V. Define the addition u+v= (x_1, y_1) +(x_2, y_2) = $(2x_1-3x_2, y_1-y_2)$. Show that V is not a vector space.	L-2	CO3	**02**
	d.	Find the linear transformation $f: R^2 \to R^2$such that $f(1,0) = (1,1)$ and $f(0,1) = (-1,2)$.	L-2	CO4	**02**
	e.	Find a least squares line for the given data points (4,3) and (3,2).	L-2	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	The probability that a person A solves the problem is 1/3, that of B is 1/2 and that of C is 3/5. If the problem is simultaneously assigned to all of them what is the probability that the problem is solved?	L-3	CO1	**05**
		OR			
	b.	From 6 positive and 8 negative numbers, 4 numbers are chosen at random (without replacement) and multiplied. What is the probability that the product is a positive number?	L-3	CO1	**05**
3.	**a.**	A random variable X has the following probability function for various values of x.	L-3	CO2	**05**

X	0	1	2	3	4	5	6	7
$p(x)$	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

i. Find the value of k.

ii. Mean and Variance.

iii. Evaluate $p(x < 6)$ and $p(x \geq 6)$.

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		OR			
	b.	A car-hire firm has two cars which it hires out day by day. The number demands for a car on each day is distributed as a Poisson's distribution with mean 1.5. Calculate the proportion of days on which i. there is no demand ii. demand is refused.	L-3	CO2	**05**

4.	a.	Show that the set B= {(1,1,0), (1,0,1), (0,1,1)} is a basis of vector space in $V_3(R)$.	L-3	CO3	**05**
		OR			
	b.	Investigate the linear dependence and independence of the vectors $v_1 = (2,-1,3,2), v_2 = (1,3,4,2)$, and $v_3 = (3,-5,2,2)$.	L-3	CO3	**05**
5.	a.	Let $S$is spanned by (1,2,2,3)and (1,3,3,2). **i.** Find a basis for $S^{\perp}$. **ii.** Can every Vin R^4written uniquely in terms of $S$and $S^{\perp}$?	L-3	CO3	**05**
		OR			
	b.	If $f: V_3(R) \to V_2(R)$ is defined by $f(x,y,z) = (x+y, y+z)$. Show that f is a linear transformation.	L-3	CO4	**05**
6.	a.	Find a least squares line for the given data points (2,1),(3,2),(4,3) and (5,2).	L-3	CO6	**05**
		OR			
	b.	Find all the eigenvalues and the corresponding eigenvectors of the matrix $A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$.	L-3	CO5	**05**
7.	a.	Find all the Eigenvalues of the matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}$.	L-3	CO5	**05**
		OR			
	b.	Find a least squares solution to $Ax = b$, where $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 5 & 3 \\ 2 & 0 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} -1 \\ 2 \\ 0 \\ 1 \\ -2 \end{bmatrix}$.	L-3	CO6	**05**
		PART-C			**6 X 10=60**
8.	a.	State and prove Baye's Theorem and hence solve the following problem. Three major parties A, B and C contending for power in the elections of a state and the probability of their winning the elections is in the ratio 1:3:5. The parties A, B and C respectively have probabilities of banning the online lottery 2/3, 1/3 and 3/5. What is the probability that there will be a ban on the online lottery in the state? What is the probability that the ban is from the party C?	L-3	CO1	**10**
		OR			

	b.	Given $P(A) = 3/4, P(B) = 1/5$ and $P(A \cap B) = 1/20$. Find $P(A \cup B), P(A \cup \overline{B}), P(\overline{A}/B), P(\overline{A}/\overline{B})$ and $P(\overline{A \cup B})$?	L-3	CO1	**10**
9.	**a.**	The number of accidents per day (x) as recorded in a textile industry over a period of 400 days is given. Fit a Poisson distribution for the data and calculate the theoretical frequencies.	L-3	CO2	**10**

x	0	1	2	3	4	5
f	173	168	37	18	3	1

OR

	b.	Write the probability density function for an exponential distribution and hence use to solve the following. The length of a telephone conversation has an exponential distribution with a mean of 3 minutes. Find the probability that a call **i.** ends in less than 3 minutes **ii.** ends in more than 5 minutes **iii.** takes between 3 and 5 minutes.	L-3	CO2	**10**
10.	**a.**	Show that the set V in a vector space over the field F mentioned under usual addition and scalar multiplication where $V = \left\{ \begin{pmatrix} x & 0 \\ z & y \end{pmatrix}, x, y, z \in R \right\}$ and $R = F$.	L-3	CO3	**10**

OR

	b.	Given $A = \begin{bmatrix} 1 & -2 & 1 & 0 & 2 \\ 1 & -1 & 4 & 1 & 3 \\ -1 & 3 & 2 & 1 & -1 \\ 2 & -3 & 5 & 1 & 5 \end{bmatrix}$, compute four fundamental vector spaces associated with matrix.	L-3	CO3	**10**
11.	**a.**	Explain the following: **i.** Reflection **ii.** Projection **iii.** Magnification **iv.** Dilation **v.** Rotation	L-3	CO3	**10**

OR

	b.	Let W be the subspace of R^4 with basis $S = \{u_1, u_2, u_3\}$ where $u_1 = (1, -2, 0, 1)$, $u_2 = (-1, 0, 0, -1)$ and $u_3 = (1, 1, 0, 0)$. Use the Gram- Schmidt process to transform S to orthonormal basis for W.	L-3	CO4	**10**

12.	**a.**	Find the eigen decomposition of the matrix $A = \begin{bmatrix} 2 & 2 & 3 \\ 1 & 2 & 1 \\ 2 & -2 & 1 \end{bmatrix}$ and hence find A^4.	L-3	CO5	**10**
		OR			
	b.	Find the singular value decomposition of the matrix $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$.	L-3	CO6	**10**
13.	**a.**	Find the singular value decomposition of the matrix $A = \begin{pmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{pmatrix}$.	L-3	CO5	**10**
		OR			
	b.	Find a quadratic least squares polynomial for the given data points $(0.5, -1.6)$, $(1, 0.4)$, $(1.5, 0.7)$, $(2, 1.8)$, $(2.5, 1.6)$, $(3, 2.2)$, $(3.5, 1.7)$, $(4, 2.2)$, $(4.5, 1.6)$ and $(5, 1.5)$.	L-3	CO6	**10**